

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Web Development Intern”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Bhagyashri- 1DS20AI016

Conducted at

Mindsoft

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Web Development Intern**” carried out by **BHAGYASHRI [1DS20AI016]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head
Eshwaran K
Team Lead
Mindsoft

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **BHAGYASHRI**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **MINDSOFT**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:22-04-2024

USN: 1DS20AI016

Place: Bangalore

NAME: Bhagyashri

OFFER LETTER

Mindsoft

Ref.: MS 2024/INT/017 Date: 01.03.2024

Internship Offer Letter

Dear Bhagyashri,

We are pleased to offer you Web Development intern for 3 Months with Mindsoft.
Your internship will start from 1st March 2024.

Once you accept this offer, the terms will be crystallized in the form of an employment agreement.

Below is the related information for your reference

1. Name of the Guide allotted : Eswaran K
2. Experience of the Guide : Ten years
3. Designation of the Guide : Project Manager
4. Contact details of the Guide : 080-41270938

We look forward to you joining us. Please do not hesitate to reach us for any information you may need.

Congratulation and Welcome to the team!

Your Sincerely


SANJEEV KUMAR
Manager Mindsoft. (ITP)
Bengaluru.




Candidate Name & Sign

mindsofttech.com
contact@mindsofttech.com

1019, Sri Venkateshwara Complex, 3rd floor, 2nd Main, Service Road (Next to New Shanti Sagar)
Vijayanagar, Bangalore - 560 040. Ph : 080-23409481, Mob. : +91-9844641410

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr Eshwaran K, Mindsoft, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

BHAGYASHRI
[1DS21AI403]

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CHAPTER 1

INTRODUCTION

In the ever-evolving landscape of digital technology, web development stands as a cornerstone, shaping the online experiences of millions worldwide. As a web development intern, my journey over the past [duration of internship] has been a dynamic exploration of this intricate realm, where theory meets practice, and creativity intertwines with technical prowess.

Throughout my tenure, I have had the privilege of delving into various facets of web development, gaining insights into fundamental principles, emerging trends, and industry best practices. From HTML and CSS to JavaScript frameworks and beyond, each day presented an opportunity to expand my knowledge base and refine my skills under the guidance of seasoned professionals.

This report serves as a reflective narrative of my experiences, achievements, and challenges encountered during my internship journey. It encapsulates the projects undertaken, the skills honed, and the invaluable lessons learned along the way. Moreover, it sheds light on the impact of this internship on my personal and professional growth, underscoring the transformative power of hands-on learning in a real-world setting.

As I navigate through the following sections, I invite you to embark on this journey with me, exploring the intricacies of web development from the lens of an intern eager to contribute, learn, and make a meaningful impact in the digital sphere.

Beyond the realm of coding and development, our interns have also explored the broader implications of their work on society and industry. By embracing emerging technologies, adopting best practices, and adhering to principles of accessibility and inclusivity, they have not only created functional websites and applications but also paved the way for innovation and positive change. Through their projects, they have demonstrated a commitment to user-centered design and a passion for leveraging technology to address real-world problems, ultimately leaving a lasting impact on the digital landscape.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

2.1 Company Profile

- ✓ Mindsoft is India's premier information enabling company.
- ✓ Provides Web Development, Custom Software Development and specialized outsourced project development.
- ✓ Mindsoft is a full-service technology consulting company.
- ✓ Mindsoft offers value-added services in key areas such as system integration, networking consultancy and a wide range of support services.
- ✓ It offers one-stop-shop convenience to its diverse customers having a diverse set of requirements.

2.2 History

- ✓ Mindsoft Technologies made the first move during 1998 in India as offshore development center.
- ✓ Built on its proven ability to develop client-driven applications for every need of the business enterprise.

2.3 Vision

Mindsoft is dedicated to create simple solutions for complex situations with the Power of imagination and implementing the best technologies.

- ✓ The global delivery model is designed in combination of Onsite, Offsite and Offshore resources, to generate cost-effective solutions to maximize customer's business performance.
- ✓ The highest quality consciousness and process driven methodology helped to strive for excellence in all activities.
- ✓ Strategic thinking is the road map necessary for attaining success

2.4 Services

Services offered by the company includes the following:

- ✓ IT Consulting.
- ✓ IT Services
- ✓ IT Infrastructure Services.
- ✓ Application development services

2.4.1 IT Consulting

Mindsoft Technologies focus on delivering the optimized solution and analogous with our client's business strategies.

- ✓ A thorough understanding of the given conception.
- ✓ Mindsoft assists clients in deploying and implementing the solutions delivered.
- ✓ Visualizing and modeling with help of IT initiative enables mindsoft in delivering the precise technological solution in the most economical manner.
- ✓ Consultants enhance the strategic accomplishments with technological assistance.
- ✓ Mindsoft's technical and business consultants are highly experienced, trained and dedicated in designing winning solutions.
- ✓ Consultants are highly motivated domain experts from a variety of industries.
- ✓ Focusing on ways enhance the existing strengths, Mindsoft provides new value added strategic solutions to enhance customer's capabilities.

2.4.2 IT Services

In today's demanding business environment which prioritizes flexibility, speed, and efficiency, a robust IT strategy is critical.

- ✓ Mindsoft helps customers achieve these qualities in their IT strategy by providing customers system integration, testing, application development and management services and solutions.
- ✓ Enterprise based products designed to help clients greatly accelerate the software development process, without sacrificing quality.
- ✓ IT services are directed towards providing design and assistance in any phase of software development or IT implementation.

2.4.3 IT Infrastructure Services

Growing application dependency and dynamism of the enterprise's business environment require organizations to derive more out of its IT investments in its endeavor to align IT to business objectives. Through a proactive monitoring and performance tuning methodology,

mindsoft ensures that your business operations are seamless.

- ✓ **Delivery Driven Methodology:** Mindsoft offers comprehensive solutions using a proven methodology and approach that are both practically feasible and economically viable.
- ✓ **Automation:** Mindsoft brings in improved operational efficiency through automation.

2.4.4 Application Development Services

Our experienced system analysts and programmers work on your project from its conceptualization through and beyond its completion and implementation.

- ✓ **Manage the project requirements** to ensure that the project remains on its intended course and within budget.
- ✓ **Bound to reduce the overall development costs.**

CHAPTER 3

SCOPE OF WORK

Objective: The primary objective of the Web Development Internship is to provide hands-on experience and training in various aspects of web development, including but not limited to frontend development, backend development, database management, and version control. The intern will work under the guidance of experienced mentors to contribute to real-world projects and gain practical skills that are essential for a career in web development.

Key Responsibilities:

Frontend Development:

- Assist in the creation of user-friendly and responsive interfaces using HTML, CSS, and JavaScript.
- Collaborate with designers to implement mockups and prototypes into functional web pages.
- Ensure cross-browser compatibility and optimize website performance for various devices.

Backend Development:

- Learn to develop server-side logic using programming languages such as Python, Node.js, or PHP.
- Work with frameworks like Express.js, Django, or Flask to build robust backend systems.
- Implement RESTful APIs and integrate third-party services to enhance website functionality.

Database Management:

- Gain proficiency in database design and management using SQL or NoSQL databases.
- Learn to write efficient database queries to retrieve and manipulate data as required by the application.
- Understand concepts such as data normalization, indexing, and transaction management.

Version Control:

- Utilize version control systems such as Git to collaborate with team members and track changes to codebase.
- Learn branching and merging strategies to effectively manage codebase updates

and resolve conflicts.

- Follow best practices for commit messages, branching workflows, and repository organization.

Learning Goals:

- Develop a solid understanding of web development principles, including frontend and backend technologies, databases, and version control.
- Gain practical experience in building functional web applications from scratch and deploying them to production environments.
- Learn to troubleshoot and debug issues encountered during development, testing, and deployment phases.
- Enhance communication and collaboration skills by working closely with mentors and team members to achieve project objectives.
- Acquire knowledge of industry standards, trends, and best practices in web development to stay abreast of advancements in the field.

Deliverables:

- Regular progress updates and status reports on assigned tasks and projects.
- Completed assignments, projects, or contributions to ongoing development efforts.
- Documentation of learned concepts, methodologies, and tools for future reference and knowledge sharing.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

1. Mastery of Frontend Technologies:

Gain proficiency in HTML, CSS, and JavaScript to create visually appealing and interactive user interfaces. Understand modern frontend frameworks such as React, Angular and their application in building dynamic web applications. Learn best practices for frontend development, including responsive design, accessibility, and browser compatibility.

2. Proficiency in Backend Development:

Acquire hands-on experience in server-side programming using languages like Python and PHP. Explore backend frameworks such as Django and scalable web applications. Learn to design and implement RESTful APIs for communication between frontend and backend components.

3. Database Management Skills:

Develop a strong foundation in database design principles and SQL querying. Gain exposure to relational databases like MySQL as well as NoSQL databases like MongoDB or Firebase. Understand data modelling, normalization, indexing, and database optimization techniques.

4. Version Control and Collaboration:

Master the use of version control systems, particularly Git, for tracking changes to codebase and collaborating with team members. Learn branching and merging strategies to manage concurrent development efforts and streamline collaboration workflows. Practice effective communication and collaboration within development teams using tools like GitHub or GitLab.

5. Problem-Solving and Debugging Skills:

Develop critical thinking and problem-solving abilities to identify and resolve issues encountered during development. Learn debugging techniques and tools to diagnose and fix errors in code efficiently. Gain experience in troubleshooting common web development challenges, such as browser compatibility issues, performance bottlenecks, and security vulnerabilities.

6. Project Management and Time Management:

Acquire project management skills by working on assigned tasks and projects within specified timelines. Learn to prioritize tasks, manage workload, and adapt to changing

project requirements. Develop organizational skills to maintain project documentation, track progress, and communicate updates effectively.

7. Continuous Learning and Professional Development:

Cultivate a growth mindset and a commitment to lifelong learning in the field of web development. Stay updated on industry trends, emerging technologies, and best practices through self-study, online resources, and participation in community events. Seek feedback from mentors and peers to identify areas for improvement and actively work towards personal and professional growth.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Technology Complexity:

- Challenge: Navigating the complexity of multiple programming languages, frameworks, and tools in web development can be overwhelming.
- Solution: Break down learning into manageable chunks, focus on mastering fundamentals first, and gradually explore advanced topics with hands-on practice and guidance from mentors.

2. Debugging Issues:

- Challenge: Debugging errors and troubleshooting issues in code can be time-consuming and frustrating, especially for beginners.
- Solution: Utilize debugging tools, reference documentation, and online resources to diagnose and resolve problems systematically, while seeking assistance from peers or mentors when needed.

3. Version Control Conflicts:

- Challenge: Managing conflicts and merge issues in version control systems like Git during collaborative development can lead to code discrepancies and project delays.
- Solution: Adopt disciplined version control practices, such as branching strategies and regular commits, communicate changes effectively with team members, and utilize tools like GitKraken or SourceTree to visualize and resolve conflicts efficiently.

4. Cross-Browser Compatibility:

- Challenge: Ensuring consistent functionality and appearance across different web browsers and devices presents a significant challenge due to varying rendering engines and capabilities.
- Solution: Test web applications on multiple browsers and devices, utilize CSS frameworks like Bootstrap for responsive design, and employ feature detection techniques and polyfills to handle browser inconsistencies.

5. Performance Optimization:

- Challenge: Optimizing website performance in terms of speed, responsiveness, and resource utilization requires a deep understanding of factors such as caching,

minification, and image optimization.

- Solution: Employ performance monitoring tools like Google PageSpeed Insights or Lighthouse to identify bottlenecks, implement best practices for optimization, and leverage content delivery networks (CDNs) for faster content delivery.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Project -1

Amazon clone

The Amazon clone using HTML and CSS is a project aimed at replicating the user interface and design of the Amazon website. It involves creating various components such as navigation bars, product cards, search bars, and footer sections using HTML for structure and CSS for styling. Through this project, learners gain hands-on experience in front-end web development by understanding layout techniques, styling elements, and creating responsive designs. This clone project serves as a practical way to reinforce HTML and CSS skills while mimicking the look and feel of a popular e-commerce platform like Amazon.

Result

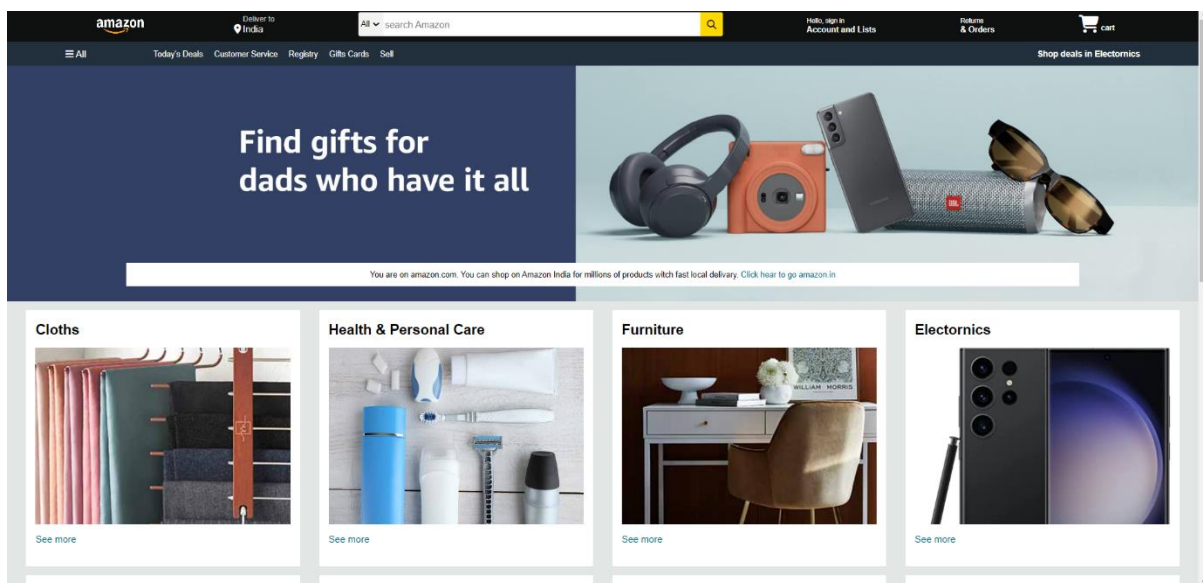


Fig 1. Amazon clone

Project – 2

CAR SEARCH ENGINE

A car search engine is a specialized online platform designed to help users find and compare different types of cars available for sale. Similar to how general search engines like Google help users find information on the internet, a car search engine focuses specifically on providing information related to automobiles.

Here's how such a project typically works:

1. **Database:** The search engine maintains a vast database of cars from various

manufacturers, dealerships, and private sellers. This database contains details such as make, model, year, mileage, price, location, and features of each car.

2. **User Interface:** Users interact with the search engine through a user-friendly interface, usually a website or a mobile app. They can input their preferences, such as the make and model of the car, price range, location, and specific features they're looking for.
3. **Search Algorithm:** The search engine employs complex algorithms to process user queries and retrieve relevant results from the database. These algorithms take into account various factors like keyword relevance, location proximity, price range, and user preferences.
4. **Filtering and Sorting:** Users can further refine their search results by applying filters based on criteria like car condition (new or used), fuel type, transmission type, body style, and more. They can also sort the results based on factors such as price, mileage, or year.
5. **Listing Details:** Each search result typically includes detailed information about the car, such as photos, specifications, seller contact information, and sometimes user reviews or ratings.
6. **Comparisons:** Some car search engines allow users to compare multiple cars side by side, enabling them to make more informed decisions based on key differences and similarities.
7. **Integration with Dealerships and Sellers:** Many car search engines partner with dealerships and individual sellers to display their inventory on the platform. Users can often contact sellers directly through the platform or visit their websites for more information or to schedule test drives.
8. **Additional Features:** Depending on the platform, there may be additional features such as loan calculators, insurance quotes, vehicle history reports, and price trend analysis to further assist users in their car buying journey.

[Home](#) [Companies](#) [Specifications](#) [Contacts](#)

Welcome to WagonAuto!!

WagonAuto is a website that gives you the details of all the cars that you want to have. We cover most common companies like Maruti Suzuki, Hyundai, Tata, Honda, Toyota, Mahindra and KIA. You can view the description of a particular car of a particular company or list the cars based on your preferences of it's specifications. Also, we have contacts and location details of several showroom that you can contact and purchase your dream car!!

The following is a guide to our website:

- HOME: Brings you to this page from wherever you are
- COMPANIES: Get all the details of a specific car
- SPECIFICATION: Get a list of cars that matches your specifications
- CONTACT: Gives you details of different showrooms

Fig 2: Index Page

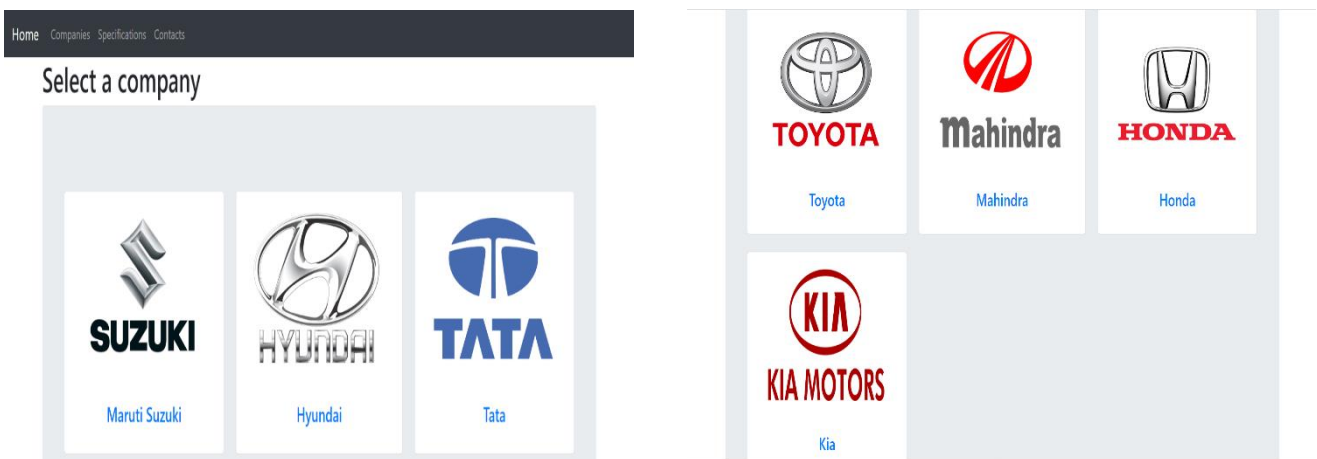


Fig 3. Companies page

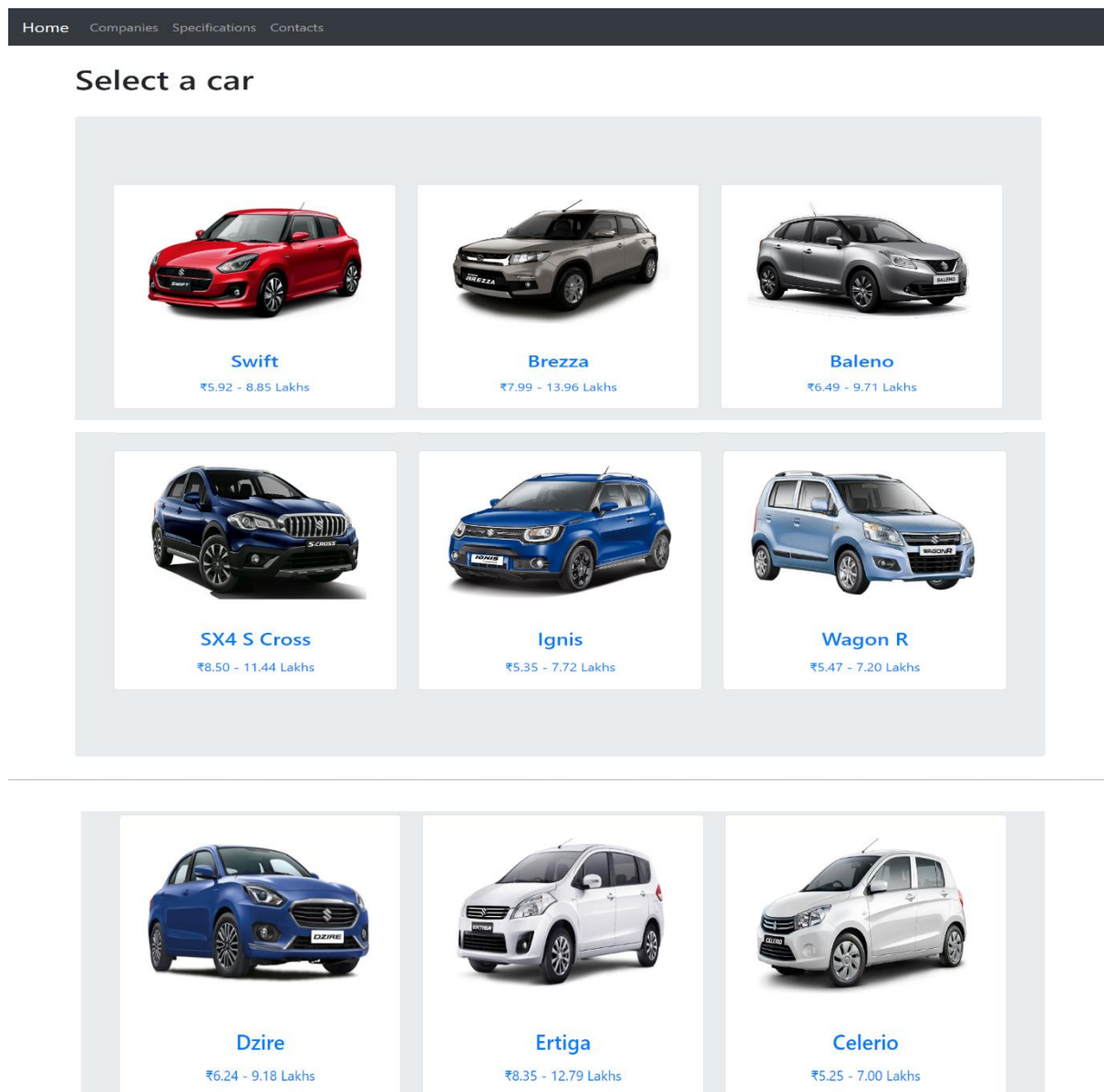


Fig 4. Page to choose car

WAGONAUTO

127.0.0.1:8000/wagonauto/suzuki/1/



Maruti Suzuki Swift

Ex-showroom price: ₹5.92 - 8.85 Lakhs

[About](#)[Engine & transmossion](#)[Fuel & Performance](#)[Dimentions & Capacity](#)[Other Comforts](#)

Maruti Suzuki Swift

The Maruti Swift has 1 Petrol Engine and 1 CNG Engine on offer. The Petrol engine is 1197 cc while the CNG engine is 1197 cc . It is available with Manual & Automatic transmission. Depending upon the variant and fuel type the Swift has a mileage of 23.2 kmpl to 30.9 km/kg . The Swift is a 5 seater 4 cylinder car and has length of 3845mm, width of 1735 and a wheelbase of 2450.

Fig 5. Car Detail page

CHAPTER 7

REFLECTION AND ANALYSIS

- ❖ **Personal Growth:** Reflecting on how the internship has contributed to personal and professional growth, including improvements in technical skills, problem-solving abilities, and confidence in tackling challenges.
- ❖ **Project Experiences:** Analysing key project experiences, highlighting successes, challenges overcome, and lessons learned throughout the development process.
- ❖ **Mentorship Impact:** Examining the impact of mentorship on learning and development, acknowledging the guidance received and the value of constructive feedback in shaping skills and perspectives.
- ❖ **Collaborative Learning:** Reflecting on the collaborative learning environment within the team, discussing the benefits of teamwork, communication, and knowledge sharing in achieving project goals.
- ❖ **Future Goals:** Outlining future goals and aspirations in the field of web development, considering areas for continued learning, skill enhancement, and professional growth beyond the internship experience.

CHAPTER 8

RECOMMENDATIONS

1. Structured Learning Path:

Design a structured learning path that covers essential concepts, technologies, and best practices in web development. Provide a curriculum or syllabus outlining the topics to be covered, along with recommended resources such as online courses, tutorials, and documentation.

2. Hands-on Projects:

Offer hands-on projects that allow interns to apply their knowledge and skills in real-world scenarios. Assign projects of varying complexity to cater to different skill levels and interests, and provide guidance and support as needed.

3. Mentorship and Guidance:

Pair each intern with a mentor who can provide guidance, support, and feedback throughout the internship. Encourage mentors to share their expertise, provide technical assistance, and offer career advice to help interns succeed.

4. Collaboration Opportunities:

Foster a collaborative environment where interns can work together on projects, share ideas, and learn from each other. Encourage teamwork, communication, and knowledge sharing through pair programming, code reviews, and collaborative problem-solving sessions.

5. Networking and Professional Development:

Facilitate networking opportunities for interns to connect with industry professionals, attend meetups or conferences, and build their professional network. Offer workshops, seminars, or guest lectures on topics related to career development, industry trends, and emerging technologies.

6. Recognition and Celebration:

Recognize and celebrate interns' achievements, milestones, and contributions throughout the internship. Showcase interns' work through presentations, demos, or portfolio showcases to highlight their skills and accomplishments.

CONCLUSION

As we come to the end of the web development internship program, it's a moment to reflect on the journey we've embarked upon together. Over the course of 3 months our interns have demonstrated remarkable dedication, enthusiasm, and growth, showcasing their potential to become future leaders in the field of web development.

Throughout the program, interns have delved deep into the intricacies of frontend and backend technologies, tackled complex projects, and collaborated effectively with peers and mentors. They have not only honed their technical skills but also developed essential soft skills such as communication, teamwork, and problem-solving, which are integral to success in the ever-evolving tech industry. As mentors, it has been inspiring to witness the passion and creativity that interns have brought to their work, as well as their resilience in the face of challenges. From building responsive user interfaces to optimizing database performance, interns have demonstrated a commitment to excellence and a hunger for continuous learning.

Looking ahead, we are confident that the skills, experiences, and connections forged during this internship will serve as a solid foundation for interns as they embark on their professional journeys. We have no doubt that they will continue to innovate, collaborate, and make meaningful contributions to the world of web development and beyond.

REFERENCES

1. MDN Web Docs - Comprehensive web development documentation.
2. Stack Overflow - Q&A community for developers.
3. W3Schools - Online tutorials and references for web development.
4. GitHub - Version control platform for collaborative coding.
5. CSS-Tricks - Articles and tutorials on CSS and frontend development.
6. Mozilla Developer Network (MDN) - Web development resources and documentation.
7. Smashing Magazine - Articles and resources on web design and development.
8. Codecademy - Interactive coding tutorials for various programming languages.
9. FreeCodeCamp - Free online coding lessons and projects.